HDOC Day-56

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Batch	15

Problem Statement

Write a C program that reads the size and elements of a variable-length integer array **arr[]**, and an integer **k** such that 0 < k <= size. Pass the array to a function named **first_negative()** using its address. Use pointers to pass the array between the calling function and **first_negative()**.

Consider every contiguous subarray of size **k** as a sliding window. Find and display the first negative integer in each window from left to right. Display **0** if a window contains no negative integer, and display **-1** if **k** is invalid. Use nested loops and separate results by spaces.

C Program

```
#include <stdio.h>

void first_negative(int *arr, int size, int k)
{
    int i, j, found;

    if (k <= 0 || k > size)
    {
        printf("-1\n");
        return;
    }

    for (i = 0; i <= size - k; i++)
    {
        found = 0;

        for (j = i; j < i + k; j++)
        {
            if (*(arr + j) < 0)
            {
                found = *(arr + j);
                break;
            }
        }

        printf("%d", found);

        if (i < size - k)
            printf(" ");
    }

    printf("\n");
}

int main()
{
    int size, i, k;

    printf("Enter size of array: ");
    scanf("%d", &size);

    int arr[size];

    printf("Enter array elements: ");
    for (i = 0; i < size; i++)
        scanf("%d", arr + i);

    printf("Enter k: ");
    scanf("%d", &k);
```

```
    printf("Output: ");
    first_negative(arr, size, k);

    return 0;
}
```

Explanation

The address of the array is passed to **first_negative()** through a pointer. The outer loop moves each sliding window from left to right. For every window, the inner loop scans its elements in order. As soon as a negative value is found, it is stored and the loop stops. If no negative value is present, **0** is printed. If **k** is invalid, the function prints **-1**.

Sample Outputs

```
Example 1
Array: 12 -1 -7 8 -15 30 16 28
k: 3
Output: -1 -1 -7 -15 -15 0

Example 2
Array: 1 2 -3 4 5 -6
k: 2
Output: 0 -3 -3 0 -6

Example 3
Array: 1 2 3 4
k: 2
Output: 0 0 0

Example 4
Array: 1 2 3
k: 4
Output: -1
```

Result

Thus, the program successfully finds the first negative integer in every sliding window of size k using pointers and nested loops, prints 0 when no negative integer exists, and -1 when k is invalid.